

Digital Electronics: Boolean Logic, Combinational Circuits & Flip-Flops

Subject: EC301 | Unit 3: Logic Design

1. Executive Overview & Syllabus Scope

Karnaugh Maps K-maps simplification, Multiplexers, Decoders, Encoders, Half/Full Adders, Latches, SR/JK/D/T Flip-Flops, and

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 3: Digital Electronics & Sequential Logic Circuits
- 1. Boolean Algebra & K-Maps: 3-variable and 4-variable Karnaugh Map minimization with Don't Care states
- 2. Combinational Circuit Design: Multiplexers 4:1, 8:1, Demultiplexers, Decoders 3:8, and Priority Encoders
- 3. Arithmetic Circuits: Half Adder, Full Adder, and Ripple Carry Adder propagation delay analysis
- 4. Sequential Logic: SR Latch, Master-Slave JK Flip-Flop race-around condition, D and T Flip-Flops
- 5. Synchronous Counter Design: Excitation tables, state diagrams, and Modulo-N counter construction

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.

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